

The Editors
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Dear Editors,

We submit the manuscript, “Reliable comparison of severe felt-intensity classifiers with versioned and dependent event data: a USGS DYFI benchmark,” for consideration as a Research Article.

Engineering model selection can be invalidated by catalogue revision, outcome-derived inputs, dependent events crossing data splits, temporal contamination, or unresolved point-score differences. The manuscript asks whether performance and model-superiority claims remain supportable when these risks are controlled. It contributes a released retrospective qualification workflow: declared candidates first pass schema, proxy, role, and temporal audits, then are scored on a fixed holdout and compared with paired sequence-cluster uncertainty.

The frozen $M \geq 5$ snapshot contains 7,520 source records, 2,362 eligible events, and a 430-event temporal holdout. The worked qualification excludes one candidate outside the declared source-only schema, supports all four admissible learned baselines over no-skill, and leaves the two lowest-Brier models unresolved. It therefore shows when a candidate is within contract and when an apparent advantage is insufficient to name a winner. Because labels are public, confirmatory reuse requires fresh hidden data or an evaluation service. The random-versus-sequence contrast is small and reversed and is reported descriptively.

The article fits the journal’s explicit scope in artificial and machine intelligence, computer science, computational methods, modelling and simulation, seismic evaluation, and engineering research methods. It follows the decision-oriented form of benchmark and engineering machine-learning comparison articles already published by the journal.

Data, code, frozen results, and reproduction instructions are available at <https://github.com/aksha74/dyfi-benchmark-reliability>. A clean clone passes the released validators and benchmark test suite and reconstructs the reported results. The scientific results remain those of immutable release v1.0.27; the JEAS manuscript adds only generated engineering-use assets derived from the same frozen inputs.

The manuscript is original and is not under consideration elsewhere. Both authors approved the manuscript and this submission. They declare no competing interests and no specific funding. The study uses aggregate public-domain event metadata, involves no human participants, and redistributes no respondent-level identifying information. Generative-AI assistance is disclosed and all assisted work was reviewed and verified by the authors. The public repository contains a non-peer-reviewed manuscript copy and source, disclosed as a preprint without a DOI; the software and derived data retain their MIT and CC0 terms.

Sincerely,

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